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Controller Order Reduction

We have shown in the previous chapters that the $\mathcal{H}_\infty$ control theory and μ synthesis can be used to design robust performance controllers for highly complex uncertain systems. However, since a great many physical plants are modeled as high order dynamical systems, the controllers designed with these methodologies typically have orders comparable to those of the plants. Simple linear controllers are normally preferred over complex linear controllers in control system designs for some obvious reasons: they are easier to understand and computationally less demanding; they are also easier to implement and have higher reliability since there are fewer things to go wrong in the hardware or bugs to fix in the software. Therefore, a lower order controller should be sought whenever the resulting performance degradation is kept within an acceptable magnitude. There are usually three ways in arriving at a lower order controller. A seemingly obvious approach is to design lower order controllers directly based on the high order models. However, this is still largely an open research problem. The Lagrange multiplier method developed in the next chapter is potentially useful for some problems. Another approach is to first reduce the order of a high order plant, and then based on the reduced plant model a lower order controller may be designed accordingly. A potential problem associated with this approach is that such a lower order controller may not even stabilize the full order plant since the error information between the full order model and the reduced order model is not considered in the design of the controller. On the other hand, one may seek to design first a high order, high performance controller and subsequently proceed with a reduction of the designed controller. This approach is usually referred to as controller reduction. A crucial consideration in controller order reduction is to take into account the closed-loop so that the closed-loop stability is guaranteed and the

performance degradation is minimized with the reduced order controllers. The purpose of this chapter is to introduce several controller reduction methods that can guarantee the closed-loop stability and possibly the closed-loop performance as well.

19.1 Controller Reduction with Stability Criteria

We consider a closed-loop system shown in Figure 19.1 where the n -th order generalized plant G is given by

$$G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right]$$

and $G_{22} = \left[\begin{array}{c|c} A & B_2 \\ \hline C_2 & D_{22} \end{array} \right]$ is a $p \times q$ transfer matrix. Suppose K is an m -th order controller which stabilizes the closed-loop system. We are interested in investigating controller reduction methods that can preserve the closed-loop stability and minimize the performance degradation of the closed-loop systems with reduced order controllers.

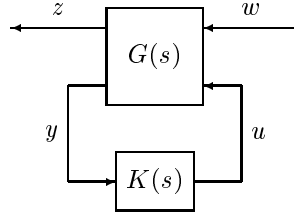


Figure 19.1: Closed-loop System Diagram

Let $\hat{K}$ be a reduced order controller and assume for the sake of argument that $\hat{K}$ has the same number of right half plane poles. Then it is obvious that the closed-loop stability is guaranteed and the closed-loop performance degradation is limited if $\|K - \hat{K}\|_\infty$ is *sufficiently* small. Hence a trivial controller reduction approach is to apply the model reduction procedure to the full order controller K . Unfortunately, this approach has only limited applications. One simple reason is that a reduced order controller that stabilizes the closed-loop system and gives satisfactory performance does not necessarily make the error $\bar{\sigma}(K - \hat{K})(j\omega)$ sufficiently small *uniformly* over all frequencies. Therefore, the approximation error only has to be made small over those critical frequency ranges that affect the closed-loop stability and performance. Since stability is the most basic requirement for a feedback system, we shall first derive controller reduction methods that guarantee this property.

19.1.1 Additive Reduction

The following lemma follows from small gain theorem.

Lemma 19.1 *Let K be a stabilizing controller and $\hat{K}$ be a reduced order controller. Suppose $\hat{K}$ and K have the same number of right half plane poles and define*

$$\Delta := \hat{K} - K, \quad W_a := (I - G_{22}K)^{-1}G_{22}.$$

Then the closed-loop system with $\hat{K}$ is stable if either

$$\|W_a\Delta\|_\infty < 1 \tag{19.1}$$

or

$$\|\Delta W_a\|_\infty < 1. \tag{19.2}$$

Proof. Since

$$I - G_{22}\hat{K} = I - G_{22}K - G_{22}\Delta = (I - G_{22}K)(I - (I - G_{22}K)^{-1}G_{22}\Delta)$$

by small gain theorem, the system is stable if $\|W_a\Delta\|_\infty < 1$. On the other hand,

$$I - \hat{K}G_{22} = I - KG_{22} - \Delta G_{22} = (I - \Delta(I - G_{22}K)^{-1}G_{22})(I - KG_{22})$$

so the system is stable if $\|\Delta W_a\|_\infty < 1$. $\square$

Now suppose K has the following state space realization

$$K = \left[\begin{array}{c|c} A_k & B_k \\ \hline C_k & D_k \end{array} \right].$$

Then

$$\begin{aligned} W_a &= \mathcal{S} \left(\left[\begin{array}{c|c} 0 & I_p \\ \hline I_q & K \end{array} \right], G_{22} \right) = \mathcal{S} \left(\left[\begin{array}{c|c|c} A_k & 0 & B_k \\ \hline 0 & 0 & I_p \\ \hline C_k & I_q & D_k \end{array} \right], \left[\begin{array}{c|c} A & B_2 \\ \hline C_2 & D_{22} \end{array} \right] \right) \\ &= \left[\begin{array}{cc|c} A + B_2 D_k R^{-1} C_2 & B_2 \tilde{R}^{-1} C_k & B_2 \tilde{R}^{-1} \\ \hline B_k R^{-1} C_2 & A_k + B_k D_{22} \tilde{R}^{-1} C_k & B_k D_{22} \tilde{R}^{-1} \\ \hline R^{-1} C_2 & R^{-1} D_{22} C_k & D_{22} \tilde{R}^{-1} \end{array} \right] \end{aligned}$$

where $R := I - D_{22}D_k$ and $\tilde{R} := I - D_k D_{22}$. Hence in general the order of W_a is equal to the sum of the orders of G and K .

In view of the above lemma, the controller K should be reduced in such a way so that the weighted error $\|W_a(K - \hat{K})\|_\infty$ or $\|(K - \hat{K})W_a\|_\infty$ is small and $\hat{K}$ and K have

the same number of unstable poles. Suppose K is unstable, then in order to make sure that $\hat{K}$ and K have the same number of right half plane poles, K is usually separated into stable and unstable parts as

$$K = K_+ + K_-$$

where K_+ is stable and a reduction is done on K_+ to obtain a reduced $\hat{K}_+$, the final reduced order controller is given by $\hat{K} = \hat{K}_+ + K_-$.

We shall illustrate the above procedure through a simple example.

Example 19.1 Consider a system with

$$A = \begin{bmatrix} -1 & 0 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad B_1 = C_1^* = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B_2 = C_2^* = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$D_{11} = 0, \quad D_{12} = D_{21}^* = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad D_{22} = 0.$$

A controller minimizing $\|T_{zw}\|_2$ is given by

$$K = -\frac{148.79(s+1)(s+3)}{(s+31.74)(s+3.85)(s-9.19)}$$

with $\|T_{zw}\|_2 = 55.09$. Since K is unstable, we need to separate K into stable part and antistable part $K = K_+ + K_-$ with

$$K_+ = -\frac{114.15(s+3.61)}{(s+31.74)(s+3.85)}, \quad K_- = \frac{-34.64}{s-9.19}.$$

Next apply the frequency weighted balanced model reduction in Chapter 7 to

$$\left\| W_a(K_+ - \hat{K}_+) \right\|_\infty,$$

we have

$$\hat{K}_+ = -\frac{117.085}{s+34.526}$$

and

$$\hat{K} := \hat{K}_+ + K_- = -\frac{151.72(s+0.788)}{(s+34.526)(s-9.19)}.$$

The $\|T_{zw}\|_2$ with the reduced order controller $\hat{K}$ is 61.69. On the other hand, if K_+ is reduced directly by balanced truncation without stability weighting W_a , then the reduced order controller does not stabilize the closed-loop system. The results are summarized in Table 19.1 where both weighted and unweighted errors are listed.

Methods	$\ K - \hat{K}\ _\infty$	$\ W_a(K - \hat{K})\ _\infty$	$\ T_{zw}\ _2$
BT	0.1622	2.5295	unstable
WBT	0.1461	0.471	61.69

Table 19.1: BT: Balance and Truncate, WBT: Weighted Balance and Truncate

It may seem somewhat strange that the unweighted error $\|K - \hat{K}\|_\infty$ resulted from weighted balanced reduction is actually smaller than that from unweighted balanced reduction. This happens because the balanced model reduction is not optimal in $\mathcal{L}_\infty$ norm. We should also point out that the stability criterion $\|W_a(K - \hat{K})\|_\infty < 1$ (or $\|(K - \hat{K})W_a\|_\infty < 1$) is only sufficient. Hence having $\|W_a(K - \hat{K})\|_\infty \geq 1$ does not necessarily imply that $\hat{K}$ is not a stabilizing controller. $\diamond$

19.1.2 Coprime Factor Reduction

It is clear that the additive controller reduction method in the last section is somewhat restrictive, in particular, this method can not be used to reduce the controller order if the controller is totally unstable, i.e., K has all poles in the right half plane. This motivates the following coprime factorization reduction approach.

Let G_{22} and K have the following left and right coprime factorizations, respectively

$$G_{22} = \tilde{M}^{-1}\tilde{N} = NM^{-1}, \quad K = \tilde{V}^{-1}\tilde{U} = UV^{-1}$$

and define

$$\begin{bmatrix} \tilde{N}_n & \tilde{M}_n \end{bmatrix} := (\tilde{M}V - \tilde{N}U)^{-1} \begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} = V^{-1}(I - G_{22}K)^{-1} \begin{bmatrix} G_{22} & I \end{bmatrix}$$

and

$$\begin{bmatrix} N_n \\ M_n \end{bmatrix} := \begin{bmatrix} N \\ M \end{bmatrix} (\tilde{V}M - \tilde{U}N)^{-1} = \begin{bmatrix} G_{22} \\ I \end{bmatrix} (I - KG_{22})^{-1}\tilde{V}^{-1}.$$

Note that $\tilde{M}_n, \tilde{N}_n, M_n, N_n$ do not depend upon the specific coprime factorizations of G_{22} .

Lemma 19.2 *Let $\hat{U}, \hat{V} \in \mathcal{RH}_\infty$ be the reduced right coprime factors of U and V . Then $\hat{K} := \hat{U}\hat{V}^{-1}$ stabilizes the system if*

$$\left\| \begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} \left(\begin{bmatrix} U \\ V \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_\infty < 1. \quad (19.3)$$

Similarly, let $\hat{\tilde{U}}, \hat{\tilde{V}} \in \mathcal{RH}_\infty$ be the reduced left coprime factors of $\tilde{U}$ and $\tilde{V}$. Then $\hat{K} := \hat{\tilde{V}}^{-1} \hat{\tilde{U}}$ stabilizes the system if

$$\left\| \left(\begin{bmatrix} \tilde{U} & \tilde{V} \end{bmatrix} - \begin{bmatrix} \hat{\tilde{U}} & \hat{\tilde{V}} \end{bmatrix} \right) \begin{bmatrix} -N_n \\ M_n \end{bmatrix} \right\|_\infty < 1. \quad (19.4)$$

Proof. We shall only show the results for the right coprime controller reduction. The case for the left coprime factorization is analogous. It is well known that $\hat{K} := \hat{U}\hat{V}^{-1}$ stabilizes the system if and only if $(\tilde{M}\hat{V} - \tilde{N}\hat{U})^{-1} \in \mathcal{RH}_\infty$. Since

$$\tilde{M}\hat{V} - \tilde{N}\hat{U} = (\tilde{M}V - \tilde{N}U) \begin{bmatrix} I - \begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} \begin{bmatrix} U - \hat{U} \\ V - \hat{V} \end{bmatrix} \end{bmatrix}$$

the stability of the closed-loop system is guaranteed if

$$\left\| \begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} \begin{bmatrix} U - \hat{U} \\ V - \hat{V} \end{bmatrix} \right\|_\infty < 1.$$

□

Now suppose $\hat{U}$ and $\hat{V}$ have the following state space realizations

$$\begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C}_1 & \hat{D}_1 \\ \hat{C}_2 & \hat{D}_2 \end{array} \right] \quad (19.5)$$

and suppose $\hat{D}_2$ is nonsingular. Then the reduced order controller is given by

$$\hat{K} = \left[\begin{array}{c|c} \hat{A} - \hat{B}\hat{D}_2^{-1}\hat{C}_2 & \hat{B}\hat{D}_2^{-1} \\ \hline \hat{C}_1 - \hat{D}_1\hat{D}_2^{-1}\hat{C}_2 & \hat{D}_1\hat{D}_2^{-1} \end{array} \right]. \quad (19.6)$$

Similarly, suppose $\hat{\tilde{U}}$ and $\hat{\tilde{V}}$ have the following state space realizations

$$\begin{bmatrix} \hat{\tilde{U}} & \hat{\tilde{V}} \end{bmatrix} = \left[\begin{array}{c|cc} \hat{\tilde{A}} & \hat{\tilde{B}}_1 & \hat{\tilde{B}}_2 \\ \hline \hat{\tilde{C}} & \hat{\tilde{D}}_1 & \hat{\tilde{D}}_2 \end{array} \right] \quad (19.7)$$

and suppose $\hat{\tilde{D}}_2$ is nonsingular. Then the reduced order controller is given by

$$\hat{K} = \left[\begin{array}{c|c} \hat{\tilde{A}} - \hat{\tilde{B}}_2\hat{\tilde{D}}_2^{-1}\hat{\tilde{C}} & \hat{\tilde{B}}_1 - \hat{\tilde{B}}_2\hat{\tilde{D}}_2^{-1}\hat{\tilde{D}}_1 \\ \hline \hat{\tilde{D}}_2^{-1}\hat{\tilde{C}} & \hat{\tilde{D}}_2^{-1}\hat{\tilde{D}}_1 \end{array} \right]. \quad (19.8)$$

It is clear from this lemma that the coprime factors of the controller should be reduced so that the weighted errors in (19.3) and (19.4) are small. Note that there is no restriction

on the right half plane poles of the controller. In particular, K and $\hat{K}$ may have different number of right half plane poles. It is also not hard to see that the additive reduction method in the last subsection may be regarded as a special case of the coprime factor reduction if the controller K is stable by taking $V = I$ or $\tilde{V} = I$.

Let L, F, L_k and F_k be any matrices such that $A + LC_2, A + B_2F, A_k + L_kC_k$ and $A_k + B_kF_k$ are stable. Define

$$\begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} = \left[\begin{array}{c|c|c} A + LC_2 & B_2 + LD_{22} & L \\ \hline C_2 & D_{22} & I \end{array} \right], \quad \begin{bmatrix} N \\ M \end{bmatrix} = \left[\begin{array}{c|c} \frac{A + B_2F}{C_2 + D_{22}F} & \frac{B_2}{D_{22}} \\ \hline F & I \end{array} \right],$$

$$\begin{bmatrix} U \\ V \end{bmatrix} = \left[\begin{array}{c|c} \frac{A_k + B_kF_k}{C_k + D_kF_k} & \frac{B_k}{D_k} \\ \hline F_k & I \end{array} \right], \quad \begin{bmatrix} \tilde{U} & \tilde{V} \end{bmatrix} = \left[\begin{array}{c|c} \frac{A_k + L_kC_k}{C_k} & \frac{B_k + L_kD_k}{D_k} \frac{L_k}{I} \\ \hline C_k & D_k \end{array} \right].$$

Then $K = UV^{-1} = \tilde{V}^{-1}\tilde{U}$ and $G_{22} = NM^{-1} = \tilde{M}^{-1}\tilde{N}$ are right and left coprime factorizations over $\mathcal{RH}_\infty$, respectively. Moreover,

$$\begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} = \left[\begin{array}{cc|cc} A + B_2D_kR^{-1}C_2 & -B_2\tilde{R}^{-1}C_k & B_2\tilde{R}^{-1} & -B_2D_kR^{-1} \\ -B_kR^{-1}C_2 & A_k + B_kR^{-1}D_{22}C_k & -B_kR^{-1}D_{22} & B_kR^{-1} \\ \hline -R^{-1}C_2 & R^{-1}D_{22}C_k - F_k & -R^{-1}D_{22} & R^{-1} \end{array} \right]$$

$$\begin{bmatrix} -N_n \\ M_n \end{bmatrix} = \left[\begin{array}{cc|c} A_k + B_kD_{22}\tilde{R}^{-1}C_k & -B_kR^{-1}C_2 & -L_k + B_kD_{22}\tilde{R}^{-1} \\ -B_2\tilde{R}^{-1}C_k & A + B_2\tilde{R}^{-1}D_kC_2 & -B_2\tilde{R}^{-1} \\ \hline -D_{22}\tilde{R}^{-1}C_k & R^{-1}C_2 & -D_{22}\tilde{R}^{-1} \\ \tilde{R}^{-1}C_k & -\tilde{R}^{-1}D_kC_2 & \tilde{R}^{-1} \end{array} \right]$$

where $R := I - D_{22}D_k$ and $\tilde{R} := I - D_kD_{22}$. Note that the orders of the weighting matrices $\begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix}$ and $\begin{bmatrix} -N_n \\ M_n \end{bmatrix}$ are in general equal to the sum of the orders of G and K . However, if K is an observer-based controller, i.e.,

$$K = \left[\begin{array}{c|c} \frac{A + B_2F + LC_2 + LD_{22}F}{F} & \frac{-L}{0} \end{array} \right],$$

letting $F_k = -(C_2 + D_{22}F)$ and $L_k = -(B_2 + LD_{22})$, we get

$$\begin{bmatrix} U \\ V \end{bmatrix} = \left[\begin{array}{c|c} \frac{A + B_2F}{F} & \frac{-L}{0} \\ \hline C_2 + D_{22}F & I \end{array} \right], \quad \begin{bmatrix} \tilde{U} & \tilde{V} \end{bmatrix} = \left[\begin{array}{c|c} \frac{A + LC_2}{F} & \frac{-L}{0} \frac{-(B_2 + LD_{22})}{I} \end{array} \right],$$

$$\tilde{M}V - \tilde{N}U = I, \quad \tilde{V}M - \tilde{U}N = I.$$

Therefore, we can chose $\begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} = \begin{bmatrix} -\tilde{N} & \tilde{M} \end{bmatrix}$ and $\begin{bmatrix} -N_n \\ M_n \end{bmatrix} = \begin{bmatrix} -N \\ M \end{bmatrix}$ which have the same orders as that of the plant G .

We shall also illustrate the above procedure through a simple example.

Example 19.2 Consider a system with

$$A = \begin{bmatrix} -1 & 0 & 4 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad B_1 = C_1^* = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B_2 = C_2^* = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$D_{11} = 0, \quad D_{12} = D_{21}^* = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad D_{22} = 0.$$

A controller minimizing $\|T_{zw}\|_2$ is given by

$$K = \left[\begin{array}{ccc|c} -1 & -8.198 & 4 & 0 \\ -8.198 & -18.396 & -8.198 & 8.198 \\ 0 & -8.198 & -3 & 0 \\ \hline 0 & -8.198 & 0 & 0 \end{array} \right] = -\frac{67.2078(s+1)(s+3)}{(s+23.969)(s+3.7685)(s-5.3414)}$$

with $\|T_{zw}\|_2 = 37.02$. Since the controller is an observer-based controller, a natural coprime factorization of K is given by

$$\begin{bmatrix} U \\ V \end{bmatrix} = \left[\begin{array}{ccc|c} -1 & -8.198 & 4 & 0 \\ 0 & -10.198 & 0 & 8.198 \\ 0 & -8.198 & -3 & 0 \\ \hline 0 & -8.198 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{array} \right].$$

Furthermore, we have

$$\begin{bmatrix} -\tilde{N} & \tilde{M} \end{bmatrix} = \left[\begin{array}{ccc|cc} -1 & 0 & 4 & -1 & 0 \\ -8.198 & -10.198 & -8.198 & -1 & -8.198 \\ 0 & 0 & -3 & -1 & 0 \\ \hline 1 & 1 & 1 & 0 & 0 \end{array} \right].$$

Applying the frequency weighted balanced model reduction in Chapter 7 to the weighted coprime factors, we obtain a first order approximation

$$\begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} = \left[\begin{array}{c|c} -0.1814 & 1.0202 \\ \hline 1.2244 & 0 \\ 6.504 & 1 \end{array} \right]$$

which gives

$$\left\| \begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} \left(\begin{bmatrix} U \\ V \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_\infty = 2.0928 > 1,$$

$$\hat{K} = \frac{1.2491}{s + 6.8165}, \quad \|T_{zw}\|_2 = 52.29$$

and a second order approximation

$$\begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} = \left[\begin{array}{cc|c} -0.1814 & 14.5511 & 1.0202 \\ -0.5288 & -4.1434 & -1.2642 \\ \hline 1.2244 & 29.5182 & 0 \\ 6.504 & -1.3084 & 1 \end{array} \right]$$

which gives

$$\left\| \begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} \left(\begin{bmatrix} U \\ V \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_\infty = 0.024 < 1,$$

$$\hat{K} = -\frac{36.069(s + 1.1102)}{(s + 17.3741)(s - 4.7601)}, \quad \|T_{zw}\|_2 = 39.14.$$

Note that the first order reduced controller does not have the same number of right half plane poles as that of the full order controller. Moreover, the sufficient stability condition is not satisfied nevertheless the controller is a stabilizing controller. It is also interesting to note that the unstable pole of the second order reduced controller is not at the same location as that of the full order controller. $\diamond$

19.2 $\mathcal{H}_\infty$ Controller Reductions

In this section, we consider $\mathcal{H}_\infty$ performance preserving controller order reduction problem. Again we consider the feedback system shown in Figure 19.1 with a generalized plant realization given by

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right].$$

The following assumptions are made:

- (A1) (A, B_2) is stabilizable and (C_2, A) is detectable;
- (A2) D_{12} has full column rank and D_{21} has full row rank;
- (A3) $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all ω ;

$$(A4) \quad \begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix} \text{ has full row rank for all } \omega.$$

It is shown in Chapter 17 that all stabilizing controllers satisfying $\|T_{zw}\|_\infty < \gamma$ can be parameterized as

$$K = \mathcal{F}_\ell(M_\infty, Q), \quad Q \in \mathcal{RH}_\infty, \quad \|Q\|_\infty < \gamma \quad (19.9)$$

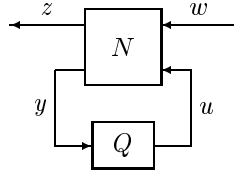
where M_∞ is of the form

$$M_\infty = \begin{bmatrix} M_{11}(s) & M_{12}(s) \\ M_{21}(s) & M_{22}(s) \end{bmatrix} = \left[\begin{array}{c|cc} \hat{A} & \hat{B}_1 & \hat{B}_2 \\ \hline \hat{C}_1 & \hat{D}_{11} & \hat{D}_{12} \\ \hat{C}_2 & \hat{D}_{21} & \hat{D}_{22} \end{array} \right]$$

such that $\hat{D}_{12}$ and $\hat{D}_{21}$ are invertible and $\hat{A} - \hat{B}_2 \hat{D}_{12}^{-1} \hat{C}_1$ and $\hat{A} - \hat{B}_1 \hat{D}_{21}^{-1} \hat{C}_2$ are both stable, i.e., M_{12}^{-1} and M_{21}^{-1} are both stable.

The problem to be considered here is to find a controller $\hat{K}$ with a minimal possible order such that the $\mathcal{H}_\infty$ performance requirement $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$ is satisfied. This is clearly equivalent to finding a Q so that it satisfies the above constraint and the order of $\hat{K}$ is minimized. Instead of choosing Q directly, we shall approach this problem from a different perspective. The following lemma is useful in the subsequent development and can be regarded as a special case of Theorem 11.7 (main loop theorem).

Lemma 19.3 *Consider a feedback system shown below*



where N is a suitably partitioned transfer matrix

$$N(s) = \begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix}.$$

Then, the closed-loop transfer matrix from w to z is given by

$$T_{zw} = \mathcal{F}_\ell(N, Q) = N_{11} + N_{12}Q(I - N_{22}Q)^{-1}N_{21}.$$

Assume that the feedback loop is well-posed, i.e., $\det(I - N_{22}(\infty)Q(\infty)) \neq 0$, and either $N_{21}(j\omega)$ has full row rank for all $\omega \in \mathbb{R} \cup \infty$ or $N_{12}(j\omega)$ has full column rank for all $\omega \in \mathbb{R} \cup \infty$ and $\|N\|_\infty \leq 1$ then $\|\mathcal{F}_\ell(N, Q)\|_\infty < 1$ if $\|Q\|_\infty < 1$.

Proof. We shall assume N_{21} has full row rank. The case when N_{12} has full column rank can be shown in the same way.

To show that $\|T_{zw}\|_\infty < 1$, consider the closed-loop system at any frequency $s = j\omega$ with the signals fixed as complex constant vectors. Let $\|Q\|_\infty =: \epsilon < 1$ and note that $T_{wy} = N_{21}^+(I - N_{22}Q)$ where N_{21}^+ is a right inverse of N_{21} . Also let $\kappa := \|T_{wy}\|_\infty$. Then $\|w\|_2 \leq \kappa\|y\|_2$, and $\|N\|_\infty \leq 1$ implies that $\|z\|_2^2 + \|y\|_2^2 \leq \|w\|_2^2 + \|u\|_2^2$. Therefore,

$$\|z\|_2^2 \leq \|w\|_2^2 + (\epsilon^2 - 1)\|y\|_2^2 \leq [1 - (1 - \epsilon^2)\kappa^{-2}]\|w\|_2^2$$

which implies $\|T_{zw}\|_\infty < 1$. $\square$

19.2.1 Additive Reduction

Consider the class of (reduced order) controllers that can be represented in the form

$$\hat{K} = K_0 + W_2\Delta W_1,$$

where K_0 may be interpreted as a nominal, higher order controller, Δ is a stable perturbation, with stable, minimum phase, and invertible weighting functions W_1 and W_2 . Suppose that $\|\mathcal{F}_\ell(G, K_0)\|_\infty < \gamma$. A natural question is whether it is possible to obtain a reduced order controller $\hat{K}$ in this class such that the $\mathcal{H}_\infty$ performance bound remains valid when $\hat{K}$ is in place of K_0 . Note that this is somewhat a special case of the above general problem; the specific form of $\hat{K}$ restricts that $\hat{K}$ and K_0 must possess the same right half plane poles, thus to a certain degree limiting the set of attainable reduced order controllers.

Suppose $\hat{K}$ is a suboptimal $\mathcal{H}_\infty$ controller, i.e., there is a $Q \in \mathcal{RH}_\infty$ with $\|Q\|_\infty < \gamma$ such that $\hat{K} = \mathcal{F}_\ell(M_\infty, Q)$. It follows from simple algebra that

$$Q = \mathcal{F}_\ell(\bar{K}_a^{-1}, \hat{K})$$

where

$$\bar{K}_a^{-1} := \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} M_\infty^{-1} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}.$$

Furthermore, it follows from straightforward manipulations that

$$\begin{aligned} \|Q\|_\infty < \gamma &\iff \left\| \mathcal{F}_\ell(\bar{K}_a^{-1}, \hat{K}) \right\|_\infty < \gamma \\ &\iff \left\| \mathcal{F}_\ell(\bar{K}_a^{-1}, K_0 + W_2\Delta W_1) \right\|_\infty < \gamma \\ &\iff \left\| \mathcal{F}_\ell(\tilde{R}, \Delta) \right\|_\infty < 1 \end{aligned}$$

where

$$\tilde{R} = \begin{bmatrix} \gamma^{-1/2}I & 0 \\ 0 & W_1 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} \gamma^{-1/2}I & 0 \\ 0 & W_2 \end{bmatrix}$$

and R is given by the star product

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} = S(\bar{K}_a^{-1}, \begin{bmatrix} K_o & I \\ I & 0 \end{bmatrix}).$$

It is easy to see that $\tilde{R}_{12}$ and $\tilde{R}_{21}$ are both minimum phase and invertible, and hence have full column and full row rank, respectively for all $\omega \in \mathbb{R} \cup \infty$. Consequently, by invoking Lemma 19.3, we conclude that if $\tilde{R}$ is a contraction and $\|\Delta\|_\infty < 1$ then $\|\mathcal{F}_\ell(\tilde{R}, \Delta)\|_\infty < 1$. This guarantees the existence of a Q such that $\|Q\|_\infty < \gamma$, or equivalently, the existence of a $\hat{K}$ such that $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$. This observation leads to the following theorem.

Theorem 19.4 *Suppose W_1 and W_2 are stable, minimum phase and invertible transfer matrices such that $\tilde{R}$ is a contraction. Let K_0 be a stabilizing controller such that $\|\mathcal{F}_\ell(G, K_0)\|_\infty < \gamma$. Then $\hat{K}$ is also a stabilizing controller such that $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$ if*

$$\|\Delta\|_\infty = \|W_2^{-1}(\hat{K} - K_0)W_1^{-1}\|_\infty < 1.$$

Since $\tilde{R}$ can always be made contractive for sufficiently small W_1 and W_2 , there are infinite many W_1 and W_2 that satisfy the conditions in the theorem. It is obvious that to make $\|W_2^{-1}(\hat{K} - K_0)W_1^{-1}\|_\infty < 1$ for some $\hat{K}$, one would like to select the “largest” W_1 and W_2 .

Lemma 19.5 *Assume $\|R_{22}\|_\infty < \gamma$ and define*

$$L = \begin{bmatrix} L_1 & L_2 \\ L_2^\sim & L_3 \end{bmatrix} = \mathcal{F}_\ell\left(\begin{array}{cc|cc} 0 & -R_{11} & 0 & R_{12} \\ -R_{11}^\sim & 0 & R_{21}^\sim & 0 \\ \hline 0 & R_{21} & 0 & -R_{22} \\ R_{12}^\sim & 0 & -R_{22}^\sim & 0 \end{array}\right), \gamma^{-1}I).$$

Then $\tilde{R}$ is a contraction if W_1 and W_2 satisfy

$$\begin{bmatrix} (W_1^\sim W_1)^{-1} & 0 \\ 0 & (W_2 W_2^\sim)^{-1} \end{bmatrix} \geq \begin{bmatrix} L_1 & L_2 \\ L_2^\sim & L_3 \end{bmatrix}.$$

Proof. See Goddard and Glover [1993]. □

An algorithm that maximizes $\det(W_1^\sim W_1) \det(W_2 W_2^\sim)$ has been developed by Goddard and Glover [1993]. The procedure below, devised directly from the above theorem, can be used to generate a required reduced order controller which will preserve the closed-loop $\mathcal{H}_\infty$ performance bound $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$.

1. Let K_0 be a full order controller such that $\|\mathcal{F}_\ell(G, K_0)\|_\infty < \gamma$;
2. Compute W_1 and W_2 so that $\tilde{R}$ is a contraction;
3. Using model reduction method to find a $\hat{K}$ so that $\left\|W_2^{-1}(\hat{K} - K_0)W_1^{-1}\right\|_\infty < 1$.

19.2.2 Coprime Factor Reduction

The $\mathcal{H}_\infty$ controller reduction problem can also be considered in the coprime factor framework. For that purpose, we need the following alternative representation of all admissible $\mathcal{H}_\infty$ controllers.

Lemma 19.6 *The family of all admissible controllers such that $\|T_{zw}\|_\infty < \gamma$ can also be written as*

$$\begin{aligned} K(s) = \mathcal{F}_\ell(M_\infty, Q) &= (\Theta_{11}Q + \Theta_{12})(\Theta_{21}Q + \Theta_{22})^{-1} := UV^{-1} \\ &= (Q\tilde{\Theta}_{12} + \tilde{\Theta}_{22})^{-1}(Q\tilde{\Theta}_{11} + \tilde{\Theta}_{21}) := \tilde{V}^{-1}\tilde{U} \end{aligned}$$

where $Q \in \mathcal{RH}_\infty$, $\|Q\|_\infty < \gamma$, and UV^{-1} and $\tilde{V}^{-1}\tilde{U}$ are respectively right and left coprime factorizations over $\mathcal{RH}_\infty$, and

$$\begin{aligned} \Theta &= \begin{bmatrix} \Theta_{11} & \Theta_{12} \\ \Theta_{21} & \Theta_{22} \end{bmatrix} = \left[\begin{array}{c|cc} \hat{A} - \hat{B}_1\hat{D}_{21}^{-1}\hat{C}_2 & \hat{B}_2 - \hat{B}_1\hat{D}_{21}^{-1}\hat{D}_{22} & \hat{B}_1\hat{D}_{21}^{-1} \\ \hline \hat{C}_1 - \hat{D}_{11}\hat{D}_{21}^{-1}\hat{C}_2 & \hat{D}_{12} - \hat{D}_{11}\hat{D}_{21}^{-1}\hat{D}_{22} & \hat{D}_{11}\hat{D}_{21}^{-1} \\ -\hat{D}_{21}^{-1}\hat{C}_2 & -\hat{D}_{21}^{-1}\hat{D}_{22} & \hat{D}_{21}^{-1} \end{array} \right] \\ \tilde{\Theta} &= \begin{bmatrix} \tilde{\Theta}_{11} & \tilde{\Theta}_{12} \\ \tilde{\Theta}_{21} & \tilde{\Theta}_{22} \end{bmatrix} = \left[\begin{array}{c|cc} \hat{A} - \hat{B}_2\hat{D}_{12}^{-1}\hat{C}_1 & \hat{B}_1 - \hat{B}_2\hat{D}_{12}^{-1}\hat{D}_{11} & -\hat{B}_2\hat{D}_{12}^{-1} \\ \hline \hat{C}_2 - \hat{D}_{22}\hat{D}_{12}^{-1}\hat{C}_1 & \hat{D}_{21} - \hat{D}_{22}\hat{D}_{12}^{-1}\hat{D}_{11} & -\hat{D}_{22}\hat{D}_{12}^{-1} \\ \hat{D}_{12}^{-1}\hat{C}_1 & \hat{D}_{12}^{-1}\hat{D}_{11} & \hat{D}_{12}^{-1} \end{array} \right] \\ \Theta^{-1} &= \left[\begin{array}{c|cc} \hat{A} - \hat{B}_2\hat{D}_{12}^{-1}\hat{C}_1 & \hat{B}_2\hat{D}_{12}^{-1} & \hat{B}_1 - \hat{B}_2\hat{D}_{12}^{-1}\hat{D}_{11} \\ \hline -\hat{D}_{12}^{-1}\hat{C}_1 & \hat{D}_{12}^{-1} & -\hat{D}_{12}^{-1}\hat{D}_{11} \\ \hat{C}_2 - \hat{D}_{22}\hat{D}_{12}^{-1}\hat{C}_1 & \hat{D}_{22}\hat{D}_{12}^{-1} & \hat{D}_{21} - \hat{D}_{22}\hat{D}_{12}^{-1}\hat{D}_{11} \end{array} \right] \\ \tilde{\Theta}^{-1} &= \left[\begin{array}{c|cc} \hat{A} - \hat{B}_1\hat{D}_{21}^{-1}\hat{C}_2 & -\hat{B}_1\hat{D}_{21}^{-1} & \hat{B}_2 - \hat{B}_1\hat{D}_{21}^{-1}\hat{D}_{22} \\ \hline \hat{D}_{21}^{-1}\hat{C}_2 & \hat{D}_{21}^{-1} & \hat{D}_{21}^{-1}\hat{D}_{22} \\ \hat{C}_1 - \hat{D}_{11}\hat{D}_{21}^{-1}\hat{C}_2 & -\hat{D}_{11}\hat{D}_{21}^{-1} & \hat{D}_{12} - \hat{D}_{11}\hat{D}_{21}^{-1}\hat{D}_{22} \end{array} \right]. \end{aligned}$$

Proof. Note that all admissible $\mathcal{H}_\infty$ controllers are given by

$$K(s) = \mathcal{F}_\ell(M_\infty, Q)$$

where $M_\infty = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$ is given in the last subsection. Next note that from Lemma 10.2, we have

$$\begin{aligned}\Theta &= \begin{bmatrix} \Theta_{11} & \Theta_{12} \\ \Theta_{21} & \Theta_{22} \end{bmatrix} = \begin{bmatrix} M_{12} - M_{11}M_{21}^{-1}M_{22} & M_{11}M_{21}^{-1} \\ -M_{21}^{-1}M_{22} & M_{21}^{-1} \end{bmatrix} \\ \tilde{\Theta} &= \begin{bmatrix} \tilde{\Theta}_{11} & \tilde{\Theta}_{12} \\ \tilde{\Theta}_{21} & \tilde{\Theta}_{22} \end{bmatrix} = \begin{bmatrix} M_{21} - M_{22}M_{12}^{-1}M_{11} & -M_{22}M_{12}^{-1} \\ M_{12}^{-1}M_{11} & M_{12}^{-1} \end{bmatrix} \\ \Theta^{-1} &= \begin{bmatrix} M_{12}^{-1} & -M_{12}^{-1}M_{11} \\ M_{22}M_{12}^{-1} & M_{21} - M_{22}M_{12}^{-1}M_{11} \end{bmatrix} \\ \tilde{\Theta}^{-1} &= \begin{bmatrix} M_{21}^{-1} & M_{21}^{-1}M_{22} \\ -M_{11}M_{21}^{-1} & M_{12} - M_{11}M_{21}^{-1}M_{22} \end{bmatrix}.\end{aligned}$$

Then the results follow from some algebra. $\square$

Theorem 19.7 *Let $K_0 = \Theta_{12}\Theta_{22}^{-1}$ be a central $\mathcal{H}_\infty$ controller such that $\|\mathcal{F}_\ell(G, K_0)\|_\infty < \gamma$ and let $\hat{U}, \hat{V} \in \mathcal{RH}_\infty$ be such that*

$$\left\| \begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix} \Theta^{-1} \left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_\infty < 1/\sqrt{2}. \quad (19.10)$$

Then $\hat{K} = \hat{U}\hat{V}^{-1}$ is also a stabilizing controller such that $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$.

Proof. Note that by Lemma 19.6, K is an admissible controller such that $\|T_{zw}\|_\infty < \gamma$ if and only if there exists a $Q \in \mathcal{RH}_\infty$ with $\|Q\|_\infty < \gamma$ such that

$$\begin{bmatrix} U \\ V \end{bmatrix} := \begin{bmatrix} \Theta_{11}Q + \Theta_{12} \\ \Theta_{21}Q + \Theta_{22} \end{bmatrix} = \Theta \begin{bmatrix} Q \\ I \end{bmatrix} \quad (19.11)$$

and

$$K = UV^{-1}.$$

Hence, to show that $\hat{K} = \hat{U}\hat{V}^{-1}$ with $\hat{U}$ and $\hat{V}$ satisfying equation (19.10) is also a stabilizing controller such that $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$, we need to show that there is another coprime factorization for $\hat{K} = UV^{-1}$ and a $Q \in \mathcal{RH}_\infty$ with $\|Q\|_\infty < \gamma$ such that equation (19.11) is satisfied.

Define

$$\Delta := \begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix} \Theta^{-1} \left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right)$$

and partition Δ as

$$\Delta := \begin{bmatrix} \Delta_U \\ \Delta_V \end{bmatrix}.$$

Then

$$\begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} = \begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \Theta \begin{bmatrix} \gamma I & 0 \\ 0 & I \end{bmatrix} \Delta = \Theta \begin{bmatrix} -\gamma \Delta_U \\ I - \Delta_V \end{bmatrix}$$

and

$$\begin{bmatrix} \hat{U}(I - \Delta_V)^{-1} \\ \hat{V}(I - \Delta_V)^{-1} \end{bmatrix} = \Theta \begin{bmatrix} -\gamma \Delta_U(I - \Delta_V)^{-1} \\ I \end{bmatrix}.$$

Define $U := \hat{U}(I - \Delta_V)^{-1}$, $V := \hat{V}(I - \Delta_V)^{-1}$ and $Q := -\gamma \Delta_U(I - \Delta_V)^{-1}$. Then UV^{-1} is another coprime factorization for $\hat{K}$. To show that $\hat{K} = UV^{-1} = \hat{U}\hat{V}^{-1}$ is a stabilizing controller such that $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$, we need to show that $\|\gamma \Delta_U(I - \Delta_V)^{-1}\|_\infty < \gamma$, or equivalently $\|\Delta_U(I - \Delta_V)^{-1}\|_\infty < 1$. Now

$$\begin{aligned} \Delta_U(I - \Delta_V)^{-1} &= \begin{bmatrix} I & 0 \end{bmatrix} \Delta \left(I - \begin{bmatrix} 0 & I \end{bmatrix} \Delta \right)^{-1} \\ &= \mathcal{F}_\ell \left(\begin{bmatrix} 0 & \begin{bmatrix} I & 0 \end{bmatrix} \\ I/\sqrt{2} & \begin{bmatrix} 0 & I/\sqrt{2} \end{bmatrix} \end{bmatrix}, \sqrt{2}\Delta \right) \end{aligned}$$

and by Lemma 19.3 $\|\Delta_U(I - \Delta_V)^{-1}\|_\infty < 1$ since

$$\begin{bmatrix} 0 & \begin{bmatrix} I & 0 \end{bmatrix} \\ I/\sqrt{2} & \begin{bmatrix} 0 & I/\sqrt{2} \end{bmatrix} \end{bmatrix}$$

is a contraction and $\|\sqrt{2}\Delta\|_\infty < 1$. □

Similarly, we have the following theorem.

Theorem 19.8 *Let $K_0 = \tilde{\Theta}_{22}^{-1} \tilde{\Theta}_{21}$ be a central $\mathcal{H}_\infty$ controller such that $\|\mathcal{F}_\ell(G, K_0)\|_\infty < \gamma$ and let $\hat{\hat{U}}, \hat{\hat{V}} \in \mathcal{RH}_\infty$ be such that*

$$\left\| \left(\begin{bmatrix} \tilde{\Theta}_{21} \\ \tilde{\Theta}_{22} \end{bmatrix} - \begin{bmatrix} \hat{\hat{U}} \\ \hat{\hat{V}} \end{bmatrix} \right) \tilde{\Theta}^{-1} \begin{bmatrix} \gamma^{-1} I & 0 \\ 0 & I \end{bmatrix} \right\|_\infty < 1/\sqrt{2}.$$

Then $\hat{K} = \hat{\hat{V}}^{-1} \hat{\hat{U}}$ is also a stabilizing controller such that $\|\mathcal{F}_\ell(G, \hat{K})\|_\infty < \gamma$.

The above two theorems show that the $\mathcal{H}_\infty$ controller reduction problem is equivalent to a frequency weighted $\mathcal{H}_\infty$ model reduction problem.

$\mathcal{H}_\infty$ Controller Reduction Procedures

- (i) Let $K_0 = \Theta_{12}\Theta_{22}^{-1} (= \tilde{\Theta}_{22}^{-1}\tilde{\Theta}_{21})$ be a suboptimal $\mathcal{H}_\infty$ central controller ($Q = 0$) such that $\|T_{zw}\|_\infty < \gamma$.
- (ii) Find a reduced order controller $\hat{K} = \hat{U}\hat{V}^{-1}$ (or $\hat{V}^{-1}\hat{U}$) such that the following frequency weighted $\mathcal{H}_\infty$ error

$$\left\| \begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix} \Theta^{-1} \left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_\infty < 1/\sqrt{2}$$

or

$$\left\| \left(\begin{bmatrix} \tilde{\Theta}_{21} \\ \tilde{\Theta}_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \tilde{\Theta}^{-1} \begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix} \right\|_\infty < 1/\sqrt{2}.$$

- (iii) The closed-loop system with the reduced order controller $\hat{K}$ is stable and the performance is maintained with the reduced order controller, i.e.,

$$\|T_{zw}\|_\infty = \|\mathcal{F}_l(G, \hat{K})\|_\infty < \gamma.$$

19.3 Frequency-Weighted $\mathcal{L}_\infty$ Norm Approximations

We have shown in the previous sections that controller reduction problems are equivalent to frequency weighted model reduction problems. To that end, the frequency weighted balanced model reduction approach in Chapter 7 can be applied. In this section, we propose another method based on the frequency weighted Hankel norm approximation method.

Theorem 19.9 *Let $W_1(s) \in \mathcal{RH}_\infty^-$ and $W_2(s) \in \mathcal{RH}_\infty^-$ with minimal state space realizations*

$$W_1(s) = \left[\begin{array}{c|c} A_{1w} & B_{1w} \\ \hline C_{1w} & D_{1w} \end{array} \right], \quad W_2(s) = \left[\begin{array}{c|c} A_{2w} & B_{2w} \\ \hline C_{2w} & D_{2w} \end{array} \right]$$

and let $G(s) \in \mathcal{RH}_\infty$. Suppose that $\hat{G}_1(s) = \left[\begin{array}{c|c} \hat{A}_1 & \hat{B}_1 \\ \hline \hat{C}_1 & \hat{D}_1 \end{array} \right] \in \mathcal{RH}_\infty$ is an r -th order optimal Hankel norm approximation of $[W_1GW_2]_+$, i.e.,

$$\hat{G}_1 = \arg \inf_{\deg Q \leq r} \|[W_1GW_2]_+ - Q\|_H$$

and assume

$$\left[\begin{array}{c|c} A_{1w} - \lambda I & B_{1w} \\ \hline C_{1w} & D_{1w} \end{array} \right], \quad \left[\begin{array}{c|c} A_{2w} - \lambda I & B_{2w} \\ \hline C_{2w} & D_{2w} \end{array} \right]$$

have respectively full row rank and full column rank for all $\lambda = \lambda_i(\hat{A}_1)$, $i = 1, \dots, r$. Then there exist matrices X, Y, Q , and Z such that

$$A_{1w}X - X\hat{A}_1 + B_{1w}Y = 0 \quad (19.12)$$

$$C_{1w}X + D_{1w}Y = \hat{C}_1 \quad (19.13)$$

$$QA_{2w} - \hat{A}_1Q + ZC_{2w} = 0 \quad (19.14)$$

$$QB_{2w} + ZD_{2w} = \hat{B}_1. \quad (19.15)$$

Furthermore, $G_r := \left[\begin{array}{c|c} \hat{A}_1 & Z \\ \hline Y & 0 \end{array} \right]$ is the frequency weighted optimal Hankel norm approximation, i.e.,

$$\inf_{\deg \hat{G} \leq r} \|W_1(G - \hat{G})W_2\|_H = \|W_1(G - G_r)W_2\|_H = \sigma_{r+1}([W_1GW_2]_+).$$

Proof. We shall assume $W_2 = I$ for simplicity. The general case can be proven similarly. Assume without loss of generality that $\hat{A}_1$ has a diagonal form

$$\hat{A}_1 = \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_r].$$

(The proof can be easily modified if $\hat{A}_1$ has a general Jordan canonical form). Partition X, Y , and $\hat{C}_1$ as

$$X = [X_1, X_2, \dots, X_r], \quad Y = [Y_1, Y_2, \dots, Y_r], \quad \hat{C}_1 = [\hat{C}_{11}, \hat{C}_{12}, \dots, \hat{C}_{1r}].$$

Then the equations (19.12) and (19.13) can be rewritten as

$$\begin{bmatrix} A_{1w} - \lambda_i I & B_{1w} \\ C_{1w} & D_{1w} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \end{bmatrix} = \begin{bmatrix} 0 \\ \hat{C}_{1i} \end{bmatrix}, \quad i = 1, 2, \dots, r.$$

By the assumption, the matrix

$$\begin{bmatrix} A_{1w} - \lambda_i I & B_{1w} \\ C_{1w} & D_{1w} \end{bmatrix}$$

has full row rank for all i and thus the existence of X and Y is guaranteed. Let

$$\hat{W}_1 = \left[\begin{array}{c|c} A_{1w} & -X\hat{B}_1 \\ \hline C_{1w} & -\hat{D}_1 \end{array} \right] \in \mathcal{RH}_\infty^-.$$

Then using equations (19.12) and (19.13), we get

$$W_1 G_r = \left[\begin{array}{cc|c} A_{1w} & B_{1w}Y & 0 \\ 0 & \hat{A}_1 & \hat{B}_1 \\ \hline C_{1w} & D_{1w}Y & 0 \end{array} \right] = \left[\begin{array}{cc|c} A_{1w} & A_{1w}X - X\hat{A}_1 + B_{1w}Y & -X\hat{B}_1 \\ 0 & \hat{A}_1 & \hat{B}_1 \\ \hline C_{1w} & C_{1w}X + D_{1w}Y & 0 \end{array} \right]$$

$$= \left[\begin{array}{cc|c} A_{1w} & 0 & -X\hat{B}_1 \\ 0 & \hat{A}_1 & \hat{B}_1 \\ \hline C_{1w} & \hat{C}_1 & 0 \end{array} \right] = \hat{W}_1 + \hat{G}_1.$$

Using this expression, we have

$$\begin{aligned} \|W_1(G - G_r)\|_H &= \left\| [W_1G]_+ + [W_1G]_- - \hat{W}_1 - \hat{G}_1 \right\|_H = \left\| [W_1G]_+ - \hat{G}_1 \right\|_H \\ &= \sigma_{r+1}([W_1G]_+) \leq \inf_{\deg \hat{G} \leq r} \left\| W_1(G - \hat{G}) \right\|_H \leq \|W_1(G - G_r)\|_H. \end{aligned}$$

□

Note that $Y = \hat{C}_1$ if $W_1 = I$, $Z = \hat{B}_1$ if $W_2 = I$, and the rank conditions in the above theorem are actually equivalent to the statements that the poles of $\hat{G}_1$ are not zeros of W_1 and W_2 . These conditions will of course be satisfied automatically if $W_1(s)$ and $W_2(s)$ have all zeros in the right half plane. Numerical experience shows that if the weighted Hankel approximation is used to obtain a $\mathcal{L}_\infty$ norm approximation, then choosing $W_1(s)$ and $W_2(s)$ to have all poles and zeros in the right half plane may reduce the $\mathcal{L}_\infty$ norm approximation error significantly.

Corollary 19.10 *Let $W_1(s) \in \mathcal{RH}_\infty^-$, $W_2(s) \in \mathcal{RH}_\infty^-$ and $G(s) \in \mathcal{RH}_\infty$. Then*

$$\inf_{\deg \hat{G} \leq r} \left\| W_1(G - \hat{G})W_2 \right\|_\infty \geq \inf_{\deg \hat{G} \leq r} \left\| W_1(G - \hat{G})W_2 \right\|_H = \sigma_{r+1}([W_1GW_2]_+).$$

The lower bound in the above corollary is not necessarily achievable. To make the ∞ -norm approximation error as small as possible, a suitable constant matrix D_r should be chosen so that $\left\| W_1(G - \hat{G} - D_r)W_2 \right\|_\infty$ is made as small as possible. This D_r can usually be obtained using any standard convex optimization algorithm. To further reduce the approximation error, the following optimization is suggested.

Weighted $\mathcal{L}_\infty$ Model Reduction Procedures

Let W_1 and W_2 be any antistable transfer matrices with all zeros in the right half plane.

(i) Let

$$\hat{G}_1 = \left[\begin{array}{c|c} \hat{A}_1 & Z \\ \hline Y & 0 \end{array} \right]$$

be a weighted optimal Hankel norm approximation of G .

(ii) Let the reduced order model $\hat{G}$ be parameterized as

$$\hat{G}(\theta) = \left[\begin{array}{c|c} \hat{A}_1 & B_\theta \\ \hline Y & D_\theta \end{array} \right], \quad \text{or} \quad \hat{G}(\theta) = \left[\begin{array}{c|c} \hat{A}_1 & Z \\ \hline C_\theta & D_\theta \end{array} \right].$$

(iii) Find C_θ (or B_θ) and D_θ from the following convex optimization:

$$\min_{\theta \in \mathbb{R}^m} \|W_1(G - \hat{G}(\theta))W_2\|_\infty.$$

It is noted that the weighted Hankel singular values can be used to predict the approximation error and hence to determine the order of the reduced model as in the unweighted Hankel approximation problem although we do not have an explicit $\mathcal{L}_\infty$ norm error bound in the weighted case.

If the given W_1 and W_2 do not have all poles and zeros in the right half plane, factorizations must be performed first to obtain the equivalent $\bar{W}_1(s)$ and $\bar{W}_2(s)$ so that $\bar{W}_1(s)$ and $\bar{W}_2(s)$ have all poles and zeros in the right half plane and

$$W_1^\sim(s)W_1(s) = \bar{W}_1^\sim(s)\bar{W}_1(s), \quad W_2(s)W_2^\sim(s) = \bar{W}_2(s)\bar{W}_2^\sim(s)$$

Then we have

$$\|W_1(G - \hat{G})W_2\|_\infty = \|\bar{W}_1(G - \hat{G})\bar{W}_2\|_\infty.$$

These factorizations can be easily done using Corollary 13.28 if W_1 and W_2 are stable and $W_1(\infty)$ and $W_2(\infty)$ have respectively full column rank and full row rank. For example,

assume $W_2 = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$ with D full row rank and $W_2(j\omega)W_2^*(j\omega) > 0$ for all

ω . Then there is a $M(s) = \left[\begin{array}{c|c} A & B_W \\ \hline C_W & (DD^*)^{1/2} \end{array} \right] \in \mathcal{RH}_\infty$ such that $M^{-1}(s) \in \mathcal{RH}_\infty$ and

$$W_2(s)W_2^\sim(s) = M^\sim(s)M(s)$$

where

$$B_W = PC^* + BD^*, \quad C_W = (DD^*)^{-1/2}(C - B_W^*X)$$

and

$$PA^* + AP + BB^* = 0$$

$$XA + A^*X + (C - B_W^*X)^*(DD^*)^{-1}(C - B_W^*X) = 0.$$

Finally, take $\bar{W}_2(s) = M^\sim(s)$. Then $\bar{W}_2(s)$ has all the poles and zeros in the right half plane and

$$\|W_1(G - \hat{G})W_2\|_\infty = \|W_1(G - \hat{G})\bar{W}_2\|_\infty.$$

In the case where W_1 and W_2 are not necessarily stable, the following procedures can be applied to accomplish this task.

Spectral Factorization Procedures

Let $W_1 \in \mathcal{L}_\infty$ and $W_2 \in \mathcal{L}_\infty$.

(i) Let $W_{1n} := W_1(-s)$ and $W_{2n} := W_2(-s)$.

- (ii) Let $W_{1n} = M_1^{-1}N_1$ and $W_{2n} = N_2M_2^{-1}$ be respectively the left and right coprime factorizations such that M_1 and M_2 are inner. (This step can be done using Theorem 13.34.)
- (iii) Perform the following spectral factorizations

$$N_1^\sim N_1 = V_1^\sim V_1, \quad N_2 N_2^\sim = V_2 V_2^\sim$$

so that V_1 and V_2 have all zeros in the left half plane. (Corollary 13.23 may be used here if $N_1(\infty)$ has full column rank and $N_2(\infty)$ has full row rank. Otherwise, the factorization in Section 6.1 may be used to factor out the undesirable poles and zeros in the weights.)

- (iv) Let $\bar{W}_1(s) = V_1(-s)$ and $\bar{W}_2(s) = V_2(-s)$.

We shall summarize the state space formulas for the above factorizations as a lemma.

Lemma 19.11 *Let $W(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{L}_\infty$ be a controllable and observable realization.*

- (a) *Suppose $W^\sim(j\omega)W(j\omega) > 0$ for all ω or $\left[\begin{array}{cc} A - j\omega & B \\ C & D \end{array} \right]$ has full column rank for all ω . Let*

$$Y = Ric \begin{bmatrix} -A^* & -C^*C \\ 0 & A \end{bmatrix} \geq 0$$

$$X = Ric \begin{bmatrix} -(A - BR^{-1}D^*C) & -BR^{-1}B^* \\ -C^*(I - DR^{-1}D^*)C & (A - BR^{-1}D^*C)^* \end{bmatrix} \geq 0$$

*with $R := D^*D > 0$. Then we have the following spectral factorization*

$$W^\sim W = \bar{W}^\sim \bar{W}$$

where $\bar{W}, \bar{W}^{-1} \in \mathcal{RH}_\infty^-$ and

$$\bar{W} = \left[\begin{array}{c|c} A + YC^*C & B + YC^*D \\ \hline R^{-1/2}(D^*C - B^*X)(I + YX)^{-1} & R^{1/2} \end{array} \right].$$

- (b) *Suppose $W(j\omega)W^\sim(j\omega) > 0$ for all ω or $\left[\begin{array}{cc} A - j\omega & B \\ C & D \end{array} \right]$ has full row rank for all ω . Let*

$$X = Ric \begin{bmatrix} -A & -BB^* \\ 0 & A^* \end{bmatrix} \geq 0$$

$$Y = Ric \begin{bmatrix} -(A - BD^* \tilde{R}^{-1} C)^* & -C^* \tilde{R}^{-1} C \\ -B(I - D^* \tilde{R}^{-1} D)B^* & (A - BD^* \tilde{R}^{-1} C) \end{bmatrix} \geq 0$$

with $\tilde{R} := DD^* > 0$. Then we have the following spectral factorization

$$WW^\sim = \bar{W}\bar{W}^\sim$$

where $\bar{W}, \bar{W}^{-1} \in \mathcal{RH}_\infty^-$ and

$$\bar{W} = \left[\begin{array}{c|c} A + BB^*X & (I + YX)^{-1}(BD^* - YC^*)\tilde{R}^{-1/2} \\ \hline C + DB^*X & \tilde{R}^{1/2} \end{array} \right].$$

19.4 An Example

We consider a four-disk control system studied by Enns [1984]. We shall set up the dynamical system in the standard linear fractional transformation form

$$\begin{aligned} \dot{x} &= Ax + B_1 w + B_2 u \\ z &= \begin{bmatrix} \sqrt{q_1} H \\ 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ I \end{bmatrix} u \\ y &= C_2 x + \begin{bmatrix} 0 & I \end{bmatrix} w \end{aligned}$$

where $q_1 = 1 \times 10^{-6}$, $q_2 = 1$ and

$$A = \begin{bmatrix} -0.161 & -6.004 & -0.58215 & -9.9835 & -0.40727 & -3.982 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad B_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$B_1 = \begin{bmatrix} \sqrt{q_2} B_2 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 0 & 0 & 0 & 0 & 0.55 & 11 & 1.32 & 18 \end{bmatrix}$$

$$C_2 = \begin{bmatrix} 0 & 0 & 6.4432 \times 10^{-3} & 2.3196 \times 10^{-3} & 7.1252 \times 10^{-2} & 1.0002 & 0.10455 & 0.99551 \end{bmatrix}.$$

The optimal $\mathcal{H}_\infty$ norm for T_{zw} is $\gamma_{opt} = 1.1272$. We choose $\gamma = 1.2$ to compute an 8th order suboptimal controller K_o . The coprime factorizations of K_o are obtained using Lemma 19.6 as $K_o = \Theta_{12}\Theta_{22}^{-1} = \tilde{\Theta}_{22}^{-1}\tilde{\Theta}_{21}$. The controller is reduced using several methods and the results are listed in Table 19.2 where the following abbreviations are made:

UWA Unweighted additive reduction:

$$\left\| K_o - \hat{K} \right\|_{\infty}$$

UWRCF Unweighted right coprime factor reduction:

$$\left\| \begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right\|_{\infty}$$

UWLCF Unweighted left coprime factor reduction:

$$\left\| \begin{bmatrix} \tilde{\Theta}_{21} & \tilde{\Theta}_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} & \hat{V} \end{bmatrix} \right\|_{\infty}$$

SWA Stability weighted additive reduction:

$$\left\| W_a(K_o - \hat{K}) \right\|_{\infty}$$

SWRCF Stability weighted right coprime factor reduction:

$$\left\| \begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} \left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_{\infty}$$

SWLCF Stability weighted left coprime factor reduction:

$$\left\| \left(\begin{bmatrix} \tilde{\Theta}_{12} & \tilde{\Theta}_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} & \hat{V} \end{bmatrix} \right) \begin{bmatrix} -N_n \\ M_n \end{bmatrix} \right\|_{\infty}$$

PWA Performance weighted additive reduction:

$$\left\| W_2^{-1}(K_o - \hat{K})W_1^{-1} \right\|_{\infty}$$

PWRCF Performance weighted right coprime factor reduction:

$$\left\| \begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix} \Theta^{-1} \left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_{\infty}$$

PWLCF Performance weighted left coprime factor reduction:

$$\left\| \left(\begin{bmatrix} \tilde{\Theta}_{12} & \tilde{\Theta}_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} & \hat{V} \end{bmatrix} \right) \tilde{\Theta}^{-1} \begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix} \right\|_{\infty}$$

B Balance reduction with or without weighting

H/O Hankel/convex optimization reduction with or without weighting

Table 19.3 lists the performance weighted right coprime factor reduction errors and their lower bounds obtained using Corollary 19.10. The ϵ in Table 19.3 is defined as

$$\epsilon := \left\| \begin{bmatrix} \gamma^{-1}I & 0 \\ 0 & I \end{bmatrix} \Theta^{-1} \left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_{\infty}.$$

By Corollary 19.10, $\epsilon \geq \sigma_{r+1}$ if the McMillan degree of $\begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix}$ is no greater than r .

Similarly, Table 19.4 lists the stability weighted right coprime factor reduction errors

Order of $\hat{K}$		7	6	5	4	3	2	1	0
PWA	B	1.196	1.196	1.199	1.197	U	4.99	U	U
	H/O	1.196	1.197	1.195	1.199	2.73	1.94	U	U
PWRCF	B	1.2	1.196	1.207	1.195	2.98	1.674	U	U
	H/O	1.196	1.198	1.196	1.199	2.036	1.981	U	U
PWLCF	B	1.197	1.196	U	1.197	U	U	U	U
	H/O	1.197	1.197	1.198	1.198	1.586	2.975	3.501	U
UWA	B	U	1.321	U	U	U	U	U	U
	H/O	23.15	U	U	U	U	U	U	U
UWRCF	B	1.198	1.196	1.199	1.196	U	U	U	U
	H/O	1.197	1.197	1.282	1.218	U	U	U	U
UWLCF	B	1.985	1.258	27.04	5.059	U	U	U	U
	H/O	5.273	U	U	U	U	U	U	U
SWA	B	1.327	1.199	2.27	1.47	23.5	U	U	U
	H/O	1.375	2.503	2.802	4.341	1.488	15.12	2.467	U
SWRCF	B	1.236	1.197	1.251	1.201	13.91	1.415	U	U
	H/O	2.401	1.893	1.612	1.388	2.622	3.527	U	U
SWLCF	B	1.417	1.217	48.04	3.031	U	U	U	U
	H/O	1.267	1.485	2.259	1.849	4.184	27.965	3.251	U

Table 19.2: $\mathcal{F}_\ell(G, \hat{K})$ with reduced order controller: U-closed-loop system is unstable

and their lower bounds obtained using Corollary 19.10. The ϵ_s and ϵ_u in Table 19.3 are defined as

$$\epsilon_s := \left\| \begin{bmatrix} -\tilde{N}_n & \tilde{M}_n \end{bmatrix} \left(\begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right) \right\|_\infty$$

$$\epsilon_u := \left\| \begin{bmatrix} \Theta_{12} \\ \Theta_{22} \end{bmatrix} - \begin{bmatrix} \hat{U} \\ \hat{V} \end{bmatrix} \right\|_\infty$$

where ϵ_u is obtained by taking the same $\hat{U}$ and $\hat{V}$ as in ϵ_s , not from the unweighted model reduction.

Table 19.2 shows that performance weighted controller reduction methods work very well. In particular, the PWRCF and PWLCF are easy to use and effective and there is in general no preference of using either the right coprime method or left coprime method. Although some unweighted reduction methods and some stability weighted reduction methods do give reasonable results in some cases, their performances are very hard to predict. What is the worst is that the small approximation error for the

reduction criterion may have little relevance to the closed-loop $\mathcal{H}_\infty$ performance. For example, Table 19.4 shows that the 7-th order weighted approximation error ϵ_s using the Hankel/Optimization method is very small, however, the $\mathcal{H}_\infty$ performance is very far away from the desired level.

Although the unweighted right coprime factor reduction method gives very good results for this example, one should not be led to conclude that the unweighted right coprime factor method will work well in general. If this is true, then one can easily conclude that the unweighted left coprime factor method will do equally well by considering a dual problem. Of course, Table 19.2 shows that this is not true because the unweighted left coprime factor method does not give good results. The only reason for the good performance of the right coprime factor method for this example is the special data structure of this example, i.e., the relationship between the B_1 matrix and B_2 matrix:

$$B_1 = \begin{bmatrix} \sqrt{q_2}B_2 & 0 \end{bmatrix}.$$

The interested reader may want to explore this special structure further.

Order of $\hat{K}$	r	7	6	5	4	3	2	1	0
Lower bounds	σ_{r+1}	0.295	0.303	0.385	0.405	0.635	0.668	0.687	0.702
ϵ	B	1.009	0.626	4.645	0.750	71.8	6.59	127.2	2.029
	H/O	0.295	0.323	0.389	0.658	0.960	1	1	1

Table 19.3: PWRCF: ϵ and lower bounds

19.5 Notes and References

The stability oriented controller reduction criterion is first proposed by Enns [1984]. The weighted and unweighted coprime factor controller reduction methods are proposed by Liu and Anderson [1986, 1990], Liu, Anderson, and Ly [1990], Anderson and Liu [1989], and Anderson [1993]. For normalized $\mathcal{H}_\infty$ controller, Mustafa and Glover [1991] have proposed a controller reduction method with a prior performance bounds. Normalized coprime factors have been used in McFarlane and Glover [1990] for controller order reductions. Lenz, Khargonekar and Doyle [1987] have also proposed another $\mathcal{H}_\infty$ controller reduction method with guaranteed performance for a class of $\mathcal{H}_\infty$ problems. The main results presented in this chapter are based on the work of Goddard and Glover [1993, 1994].

We should note that a satisfactory solution to the general frequency weighted $\mathcal{L}_\infty$ norm model reduction problem remains unavailable and this problem has a crucial implication toward controller reduction with preserving closed-loop $\mathcal{H}_\infty$ performance as its objective. The frequency weighted Hankel norm approximation is considered in Latham and Anderson [1986], Hung and Glover [1986], and Zhou [1993]. The $\mathcal{L}_\infty$ model reduction procedures discussed in this chapter are due to Zhou [1993].

Order of $\hat{K}$	r	7	6	5	4
Lower bounds of ϵ_s	σ_{r+1}	1.1×10^{-6}	1.2×10^{-6}	1.9×10^{-6}	1.9×10^{-6}
ϵ_s	B	0.8421	0.5048	2.5439	0.5473
	H/O	0.0001	0.2092	0.3182	0.3755
ϵ_u	B	254.28	9.7018	910.01	21.444
	H/O	185.9	30.85	305.3	15.38
Order of $\hat{K}$	r	3	2	1	0
Lower bounds of ϵ_s	σ_{r+1}	9×10^{-6}	6.23×10^{-5}	1.66×10^{-4}	2.145×10^{-4}
ϵ_s	B	11.791	1.3164	9.1461	1.5341
	H/O	0.5403	0.7642	1	1
ϵ_u	B	2600.2	365.45	3000.6	383.277
	H/O	397.9	288.1	384.3	384.3

Table 19.4: SWRCF: ϵ_s and the corresponding ϵ_u